How-To Library

Informatica

Configuring HTTPS for the Administrator Tool Using a Certificate Signed by a Certificate Authority

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Abstract

To use the SSL protocol for a secure connection to the Administrator tool, create a keystore file and configure an HTTPS port for each gateway node in the domain. The keystore file can include a self-signed certificate or a certificate signed by a certificate authority. This article describes how to create and use a certificate signed by a certificate authority.

Supported Versions

- Informatica Data Quality 9.0 - 9.1.0
- Informatica Data Services 9.0 - 9.1.0
- PowerCenter 9.0 - 9.1.0

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Overview

To use the SSL protocol for a secure connection to the Administrator tool, create a keystore file and configure an HTTPS port for each gateway node in the domain.

Create a keystore file with keytool. Keytool is a utility that generates and stores private or public key pairs and associated certificates in a file called a keystore. When you generate a public or private key pair, keytool wraps the public key into a self-signed certificate. You can use the self-signed certificate in a development or test environment. In production, use a certificate signed by a certificate authority.

If the domain contains multiple gateway nodes, you can copy the same keystore file to each gateway node. Or, if you require a unique certificate for each gateway node, create a separate keystore file for each gateway node and import the unique certificate into each keystore file.

Use the keytool utility available in Java version 1.6 or later.

Complete the following steps to configure HTTPS for the Administrator tool using a certificate signed by a certificate authority:

1. Create the keystore file.
2. Generate a certificate signing request.
3. Import the signed certificate.
4. Configure HTTPS for each gateway node in the domain.
5. Clear the browser cache and enable SSL for the browser.

This article presents a typical procedure for creating private and public key pairs and associated certificates. You may need to change the procedure to comply with the requirements of your organization. For more information about using

keytool, see the documentation on the following web site:
http://docs.oracle.com/javase/7/docs/technotes/tools/windows/keytool.html.

Step 1. Create the Keystore File

To create a keystore file, use the keytool utility available in Java version 1.6 or later.

1. Locate the keytool utility in one of the following directories:
   - `%JAVA_HOME%\jre\bin`
   - `Java\bin` directory of the Informatica Services installation directory
2. Run the keytool utility with the genkey option using the following syntax:

   ```
   keytool -genkey -alias <KeystoreAlias> -keyalg RSA -keysize 2048 -keystore
   <KeystorePathFileName>
   ```

   For example, run the following command:

   ```
   keytool -genkey -alias admintool  -keyalg RSA -keysize 2048 -keystore admintool.keystore
   ```
3. Enter the keystore password, organizational information, and key password as prompted by the utility.

   For the CN option, enter the host name of the gateway node. The example in this article uses the host name cax164796.

   After you enter all information, keytool creates a keystore file with the name and location that you specify.
4. To view the contents of the keystore file, run the keytool utility with the list option using the following syntax:

   ```
   keytool -list -v -alias <KeystoreAlias> -keystore <KeystorePathFileName>
   ```

   For example, run the following command:

   ```
   keytool -list -v -alias admintool -keystore admintool.keystore
   ```

Step 2. Generate a Certificate Signing Request

To create a certificate signed by a certificate authority, you must generate a certificate signing request (CSR) and send the CSR to a certificate authority.

1. To generate a CSR, run the keytool utility with the certreq option using the following syntax:

   ```
   keytool -certreq -keyalg RSA -alias <KeystoreAlias> -file <CSRPathFileName> -keystore
   <KeystorePathFileName>
   ```

   Use the same keystore alias and file name that you used with the genkey option.

   For example, run the following command:

   ```
   keytool -certreq -keyalg RSA -alias admintool -file admintool.csr -keystore
   admintool.keystore
   ```
2. Submit the text from the generated CSR to a certificate authority to get a digital certificate.

Step 3. Import the Signed Certificate

When you receive the signed digital certificate from the certificate authority, you must import the certificate into the keystore file that you created.

Some certificate authorities issue root and intermediate certificates that you also import into the keystore file. This step includes examples for SSL certificates provided by Go Daddy. Apply similar procedures for the certificate authority that you use.

If you request a signed certificate from Go Daddy, you receive the following certificate files that you import into the keystore file:

- valicert_class2_root.cer. The server root certificate.
- gd_cross_intermediate.crt. The intermediate cross certificate.
- gd_intermediate.crt. The intermediate certificate.
- <host_name>.crt. The signed site certificate that you requested.

Note: You also receive the gd_bundle.crt file that is not used in this example.

1. To import the root, intermediate cross, and intermediate certificates into the keystore, run the keytool utility with the import option.

   Import each certificate with a separate command. Use a unique alias name for each certificate. Use the keystore file name that you used with the genkey and certreq options.

   To import the root certificate, use the following example syntax:

   ```
   keytool -import -alias root -keystore admintool.keystore -trustcacerts -file
   valicert_class2_root.cer
   ```

   To import the intermediate cross certificate, use the following example syntax:

   ```
   keytool -import -v -alias cross -keystore admintool.keystore -trustcacerts -file
   gd_cross_intermediate.crt
   ```

   To import the intermediate certificate, use the following example syntax:

   ```
   keytool -import -alias intermed -keystore admintool.keystore -trustcacerts -file
   gd_intermediate.crt
   ```

2. To import the signed site certificate, run the keytool utility with the import option. Use the keystore alias and file name that you used with the genkey and certreq options.

   For example, use the following syntax:

   ```
   keytool -import -alias admintool -keystore admintool.keystore -trustcacerts -file
   cax164796.crt
   ```

Step 4. Configure HTTPS for Each Gateway Node in the Domain

To configure the Administrator tool to use the keystore file signed by a certificate authority, run the infasetup command line program to configure HTTPS for each gateway node in the domain.

1. Shut down the gateway node.

   If this is the only gateway node in the domain, shut down all worker nodes before you shut down the gateway node.

2. To enable HTTPS support for the gateway node, run the infasetup UpdateGatewayNode command. Include the HttpsPort, KeystoreFile, and KeystorePass options using the following syntax:

   ```
   infasetup UpdateGatewayNode -DatabaseAddress <database_hostname:database_port> -
   DatabaseUserName <database_user_name> -DatabasePassword <database_password> -
   DatabaseType <database_type> -DatabaseServiceName <database_service_name> -DomainName
   <domain_name> -HttpsPort <https_port> -KeystoreFile <keystore_file_location> -
   KeystorePass <keystore_password>
   ```

   Specify the name and location of the keystore file that you created for this gateway node. For example, run the following command:

   ```
   infasetup UpdateGatewayNode -DatabaseAddress calvin:1521 -DatabaseUserName Admin -
   DatabasePassword AdminPass -DatabaseType Oracle -DatabaseServiceName ORCL -DomainName
   ProductionDomain -HttpsPort 9091 -KeystoreFile admintool.keystore -KeystorePass changeit
   ```

3. Start the gateway node.

Step 5. Clear the Cache and Enable SSL for the Browser

Before you log in to the Administrator tool, clear the cache from the web browser and enable the browser to use SSL.

1. Launch a new browser window.
2. Clear the cookies, temporary files, and history information in the browser.
   - On Microsoft Internet Explorer, click **Tools** > **Internet Options**. On the General tab, delete the browsing history. On the Content tab, clear the SSL state.
   - On Mozilla Firefox, click **Tools** > **Clear Recent History**.
3. Enable the browser to send and receive secured information using SSL 3.0 and TLS 1.0.
   - On Microsoft Internet Explorer, click **Tools** > **Internet Options**. On the Advanced tab, select **Use SSL 3.0** and **Use TLS 1.0**.
   - On Mozilla Firefox, click **Tools** > **Internet Options**. On the Advanced tab, click **Encryption**, and then select **Use SSL 3.0** and **Use TLS 1.0**
4. Log in to the Administrator tool.

   In version 9.0, the Administrator tool URL redirects to the following HTTPS enabled site:

   ```
   https://<host>:<https port>/adminconsole
   ```

   In versions 9.0.1 and later, the Administrator tool URL redirects to the following HTTPS enabled site:

   ```
   https://<host>:<https port>/administrator
   ```
5. Click **Yes** to trust the security certificate.

Troubleshooting

If you cannot log in to the Administrator tool after configuring the tool to use HTTPS, open the web.xml file on each node in the Informatica domain. Verify that it includes the security constraint tag. The location of the file depends on the following Informatica versions:

- In version 9.0, find the file in the following location:

  ```
  <InstallationDir>\services\AdministratorConsole\adminconsole\WEB-INF
  ```
- In versions 9.0.1 and later, find the file in the following location:

  ```
  <InstallationDir>\services\AdministratorConsole\administrator\WEB-INF
  ```

Verify that the following security constraint tag is included at the end of the web.xml file:

```
<security-constraint>
   <web-resource-collection>
      <web-resource-name>Entire Application</web-resource-name>
      <url-pattern>/*</url-pattern>
   </web-resource-collection>
   <user-data-constraint>
     <transport-guarantee>CONFIDENTIAL</transport-guarantee>
   </user-data-constraint>
</security-constraint>
```

If the tag does not exist in the file, shut down the node. Add the tag to the file before the closing `</web-app>` tag, and save the file. Restart the node.

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